From Smart Technology to Smart Experience

Traditional approach

User searches → Finds information → Accesses resource

Smart experience approach

User needs → Contextual discovery → Relevant recommendations → Accessible interaction → Feedback

Key takeaway: The goal is not simply to make libraries more technological, but to make knowledge more dis and meaningful.

Digital Library vs Smart Library

Dimension	Digital Library	Smart Library
Main focus	Digital access	Intelligent services
Discovery	Search and browse	Context-aware discovery
Interaction	User-initiated	Interactive and adaptive
Personalisation	Often limited	Potentially personalised
Data	Records and usage	Data-informed decisions
Experience	Access to content	Integrated knowledge ex

A smart library builds on digital foundations. It is not a full replacement.

Technologies Driving Library Transformation

Enablers — valued for their application and benefit to users.

01 Artificial Intelligence
02 Data Analytics
03 Internet of Things
04 Cloud Computing
05 Mobile Technologies

06. AR / VR / MR

The Smart Library Ecosystem



Users
Students, researchers, lecturers, librarians and communities.

Services
Discovery, borrowing, research assistance, learning spaces.

Technology
AI, IoT, cloud, mobile applications, immersive syste

Data
Collections, metadata, usage patterns and feedbac

Governance
Privacy, accessibility, security and responsible AI.

Digital transformation is more than technology.

Digital transformation involves redesigning services, processes and experiences using digital capabilities.

Technology alone does not create transformation. Meaningful transformation happens when technology improves how people access and use knowledge.

You are viewing INTAN YUSRINA ZAIRON's screen REC View Options

he Evolution of Libraries

ceptual evolution — these models can coexist within the same institution.

01 Traditional	02 Automated	03 Digital	04 Smart
sical books and manual services	Digital catalogues and circulation	Online resources and remote access	Connected, adap intelligent ser

y message: Libraries have evolved from information storage spaces into connected knowledge and learning environ

What Is a Smart Library?

A smart library is a library environment that integrates digital technologies, connected systems, data and user-centred services to improve access to information and knowledge experiences.

01 Connected
02 Intelligent
03 Personalised
04 Accessible

05. interactive
06. data-informed

Why Must Libraries Transform?

Five major drivers of change

Changing user expectations
Fast, convenient and personalised services.

Information overload
Increasing difficulty in identifying trustworthy information.

Digital learning
Growing demand for remote and flexible access.

Emerging technologies
New possibilities for automation, interaction and discovery.

Inclusivity
The need to support diverse abilities and learning preferences.

Discussion: "If students can search for almost anything on Google or ask AI, why do they still need libraries?"

Internet of Things (IoT) in Smart Libraries

01 RFID-enabled circulation
02 Smart occupancy sensors
03 Environmental monitoring
04 Automated inventory support
05 Smart lighting and energy management

Benefits: Operational efficiency, convenience and better space management.

AI-Powered Library Assistants

"Most students must use AI for classwork, but universities are slow to offer AI courses" — eCampus News

01 24/7 basic enquiry support
02 Natural-language questions
03 Library service navigation
04 Resource discovery assistance
05 Referral to human librarians

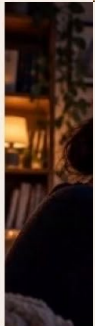
AI should complement librarians, not replace professional expertise. Example: UC Irvine Libraries offers ANTswers AI for library enquiries, with more complex questions referred to

Cloud Computing and Digital Access

Cloud-enabled library services can support:

01 Scalable digital infrastructure
02 Remote access to hosted services
03 Shared repositories
04 System integration
05 Backup and disaster recovery
06 Collaboration between institutions

Example scenario: Students access authorised digital resources from campus, home or during international mobility programmes.



Generative AI and Intelligent Search

From keyword search to conversational discovery

Traditional search

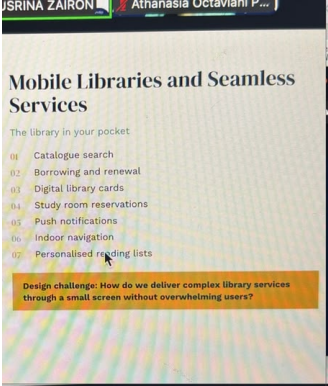
"digital literacy" AND "higher education"

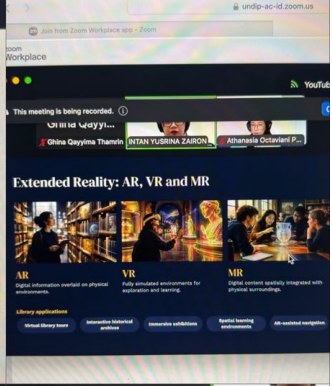
AI-assisted search

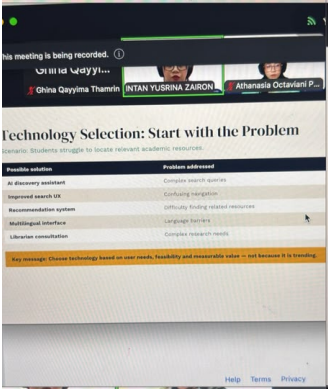
"Find recent academic resources discussing digital literacy challenges among university students, and explain how the topics are related."

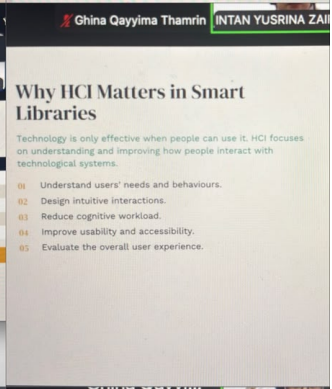
Key concepts

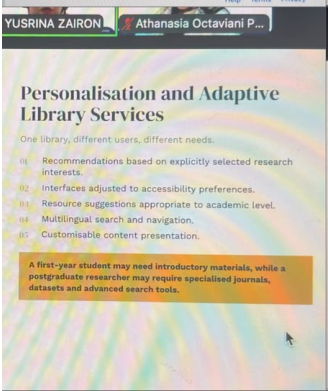
Natural Language Processing (NLP) | Semantic search | Retrieval-Augmented Generation (RAG) | Source attribution | Human verification

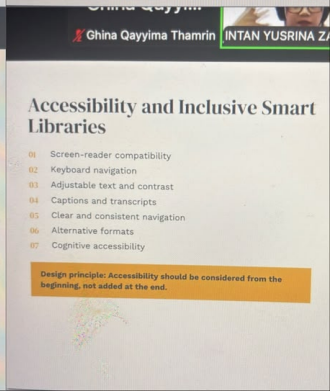

Athanasia Octaviani P...
Mobile Libraries and Seamless Services
The library in your pocket
01 Catalogue search
02 Borrowing and renewal
03 Digital library cards
04 Study room reservations
05 Push notifications
06 Indoor navigation
07 Personalised reading lists
Design challenge: How do we deliver complex library services through a small screen without overwhelming users?


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Workplace
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Ghina Qayyima Thamrin
INTAN YUSRINA ZAIRON
Athanasia Octaviani P...
Extended Reality: AR, VR and MR
AR
Digital information overlaid on physical environments.
VR
Fully simulated environments for exploration and learning.
MR
Digital content spatially integrated with physical surroundings.
Library applications
Virtual library tours
Interactive historical archives
Immersive exhibitions
Spatial learning environments
AR-assisted navigation


This meeting is being recorded.
Ghina Qayyima Thamrin
INTAN YUSRINA ZAIRON
Athanasia Octaviani P...
Technology Selection: Start with the Problem
Scenario: Students struggle to locate relevant academic resources.
Possible solution
Problem addressed
AI discovery assistant
Complex search queries
Improved search UX
Confusing navigation
Recommendation system
Difficulty finding related resources
Multilingual interface
Language barriers
Librarian consultation
Complex research needs
Key message: Choose technology based on user needs, feasibility and measurable value — not because it is trending.
Help
Terms
Privacy


Ghina Qayyima Thamrin
INTAN YUSRINA ZAI
Why HCI Matters in Smart Libraries
Technology is only effective when people can use it. HCI focuses on understanding and improving how people interact with technological systems.
01 Understand users' needs and behaviours.
02 Design intuitive interactions.
03 Reduce cognitive workload.
04 Improve usability and accessibility.
05 Evaluate the overall user experience.


YUSRINA ZAIRON
Athanasia Octaviani P...
Personalisation and Adaptive Library Services
One library, different users, different needs.
01 Recommendations based on explicitly selected research interests.
02 Interfaces adjusted to accessibility preferences.
03 Resource suggestions appropriate to academic level.
04 Multilingual search and navigation.
05 Customisable content presentation.
A first-year student may need introductory materials, while a postgraduate researcher may require specialised journals, datasets and advanced search tools.


Ghina Qayyima Thamrin
INTAN YUSRINA ZA
Accessibility and Inclusive Smart Libraries
01 Screen-reader compatibility
02 Keyboard navigation
03 Adjustable text and contrast
04 Captions and transcripts
05 Clear and consistent navigation
06 Alternative formats
07 Cognitive accessibility
Design principle: Accessibility should be considered from the beginning, not added at the end.

Zoom Workplace Meeting View Edit Window Help

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Ghina Qayyim Tham... INTAN YUSRINA ZAIRON Athanasia Octaviani P...

From Information Access to Knowledge Experienc

Traditional interaction

Search → Read → Understand

Enhanced digital experience

Discover → Explore → Interact → Reflect → Learn

Opportunities for libraries

Interactive storytelling | Multimedia-rich collections | Virtual exhibitions | Immersive environments | Collaborative expl

More

YUSRINA ZAIRON Athanasia Octaviani P...

Virtual Libraries and Digital Heritage

01 Virtual walkthroughs of library spaces.
02 Digital reconstruction of historical environments.
03 Exploration of rare manuscripts and cultural collections.
04 Interactive archival exhibitions.
05 Remote cultural heritage learning.

Example scenario: A historical manuscript can be presented alongside interactive maps, narrated explanations, timelines and 3D representations of related artefacts.

YouTube

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Ghina Qayyim Tham... INTAN YUSRINA ZAIRON Athanasia Octaviani P...

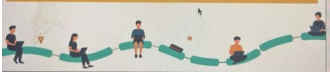
Adaptive and Personalised Learning Experiences

Concept: An adaptive digital literacy learning system.

User Profile → Learning Activity → User Interaction → Adaptation → Feedback

Identify learners' knowledge and preferences. | Offer suitable information literacy activities. | Observe progress and responses. | Adjust content difficulty or interaction. | Provide meaningful feedback.

Illustrative adaptive learning workflow, not an existing deployed library system.



More

Uploads

Tools

Photos

Projects

Ghina Qayyima Thamrin INTAN YUSRINA ZA

Research Showcase: METTAREKA and Virtual Heritage

METTAREKA — Interactive Virtual Heritage Experience: a research-based example of immersive, interactive multimedia for digital heritage.

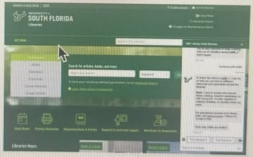
01 Virtual heritage environment
02 360° visual exploration
03 Interactive cultural information
04 Multimedia narration
05 User interaction and experience evaluation

Potential relevance: Similar design principles could support digital heritage collections, virtual exhibitions and interactive archival learning.

Case Study 1: AI-Powered Academic Library Assistance

University of South Florida Libraries — LINK AI Chatbot

01 Introduced in 2025.
02 Provides 24/7 support for library-related enquiries.
03 Helps users navigate library resources and services.
04 Supports escalation to human librarians.
05 Developed with user input and testing.



Uploads